

Causal ML as a Thinking Workflow

A step-by-step (no maths, no codes) upgrade for regression-only management researchers

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@Tongji U: Workshop on Causal Machine Learning for Management Research

Pre-seminar survey: choose your role

Role	You usually contribute
AI-assisted theorist	propose mechanisms
Regression-only econometrician	clarify estimand
Machine-learning programmer	use flexible models

The goal is not a better p-value

- | Ask ML to make the 'invisible' causal comparison visible,
- | NOT to make the 'insignificant' coefficient significant.

The current workflow feels productive

Run regression: $Y \sim X$

→

Add or delete controls

→

Keep the version where X becomes significant

→

Search interactions and name a moderator

The same coefficient hides different problems

What changed?

What might really be happening?

Coefficient moves after controls

selection, bad controls, or functional form

p-value appears after search

overfitting or specification mining

interaction is significant

real heterogeneity or chance discovery

estimate is precise

strong support or precise extrapolation

What do you want to do? Really?

The replacement workflow has four gates

Gate	Question before moving on
1. Estimand	What effect are we trying to estimate?
2. Selection	Why are treated and untreated units comparable?
3. Support and timing	Are comparisons supported, and are controls allowed?
4. Heterogeneity	Is the moderator pattern validated?

What is your workflow?

The rhythm is the same each time

Step	What we ask
1. behind the Data	What creates the next variable?
2. Old workflow	What would a regression-only researcher conclude?
3. Causal ML move	What extra diagnostic or score do we learn?
4. Decision	What should change before writing the claim?

The code is used to reveal logic

We will NOT start with software commands.

We will ask:

- What creates treatment assignment?
- What creates outcome risk?
- What does the researcher observe?
- What does the old regression miss?
- What does the ML diagnostic reveal?

(I will give you downloadable R codes to play with)

Workflow as 'stage-gates'

Gate	Student script
Control intuition	<code>01_residualization_foundation.R</code>
Observed selection	<code>02_observed_selection_dml_aipw.R</code>
Hidden confounding	<code>03_hidden_confounding_sensitivity.R</code>
Overlap	<code>04_overlap_support.R</code>
Bad controls	<code>05_bad_controls_estimand.R</code>
Heterogeneity	<code>06_hte_moderators.R</code>

Download the student R code

yangphd.com/causal_ml/R_code.zip

Run the scripts in the same order as the gates.

Foundation

RESIDUALIZATION

What does "control for W" actually do?

A small DGP (data-generating process) is the easiest

$W = 0 \text{ or } 1$

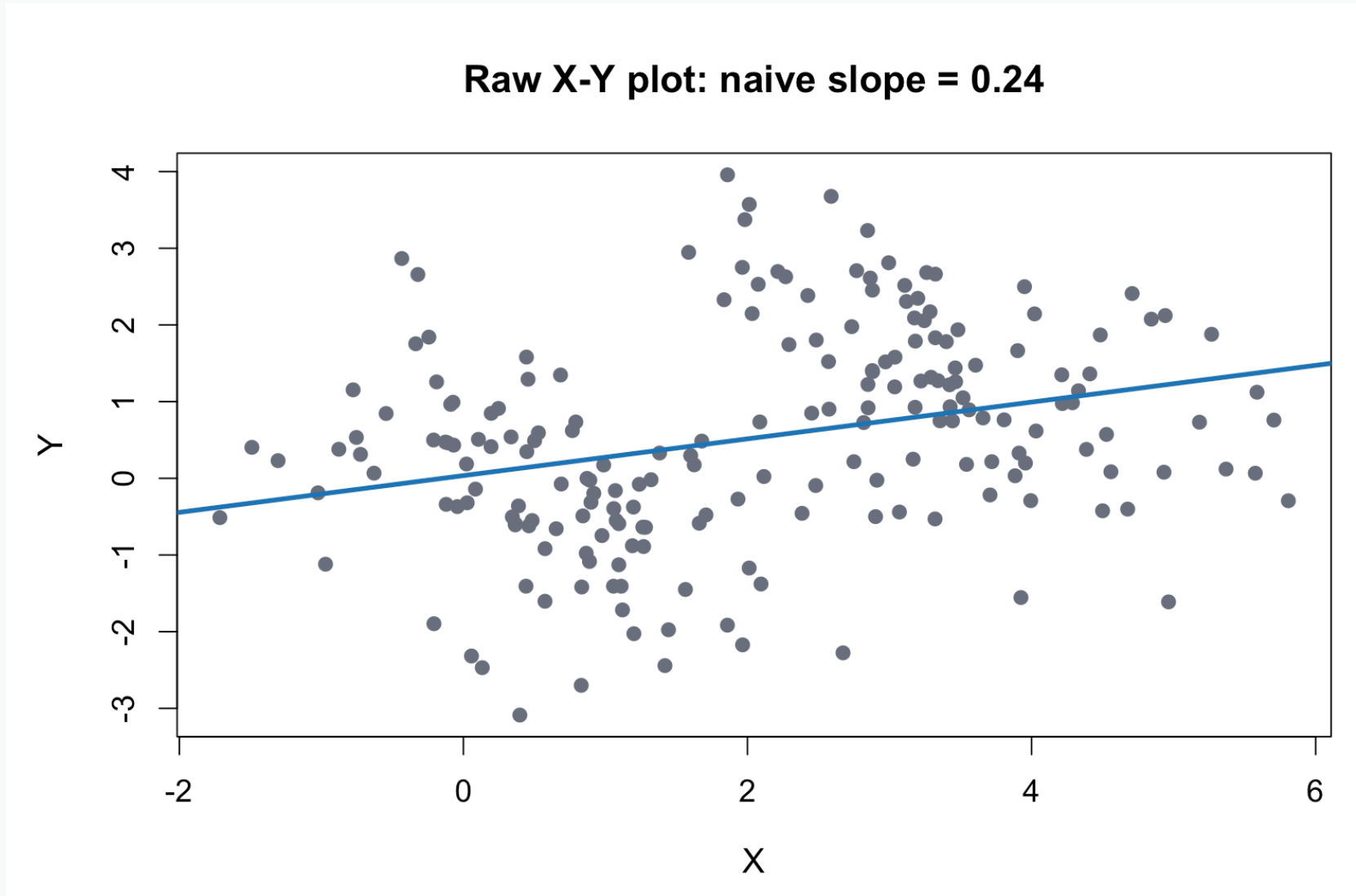
$W \rightarrow X = 0.5 + 3W + \text{noise}_X$

$W \text{ and } X \rightarrow Y = -0.5X + 3W + \text{noise}_Y$

True effect of X on Y : -0.5

DGP = what's 'behind' the data.

What you 'see' when W is 'invisible.'



Read the raw slope cautiously

W is a grouping variable.

Because **W** relates to both **X** and **Y**, the raw plot can slope upward.

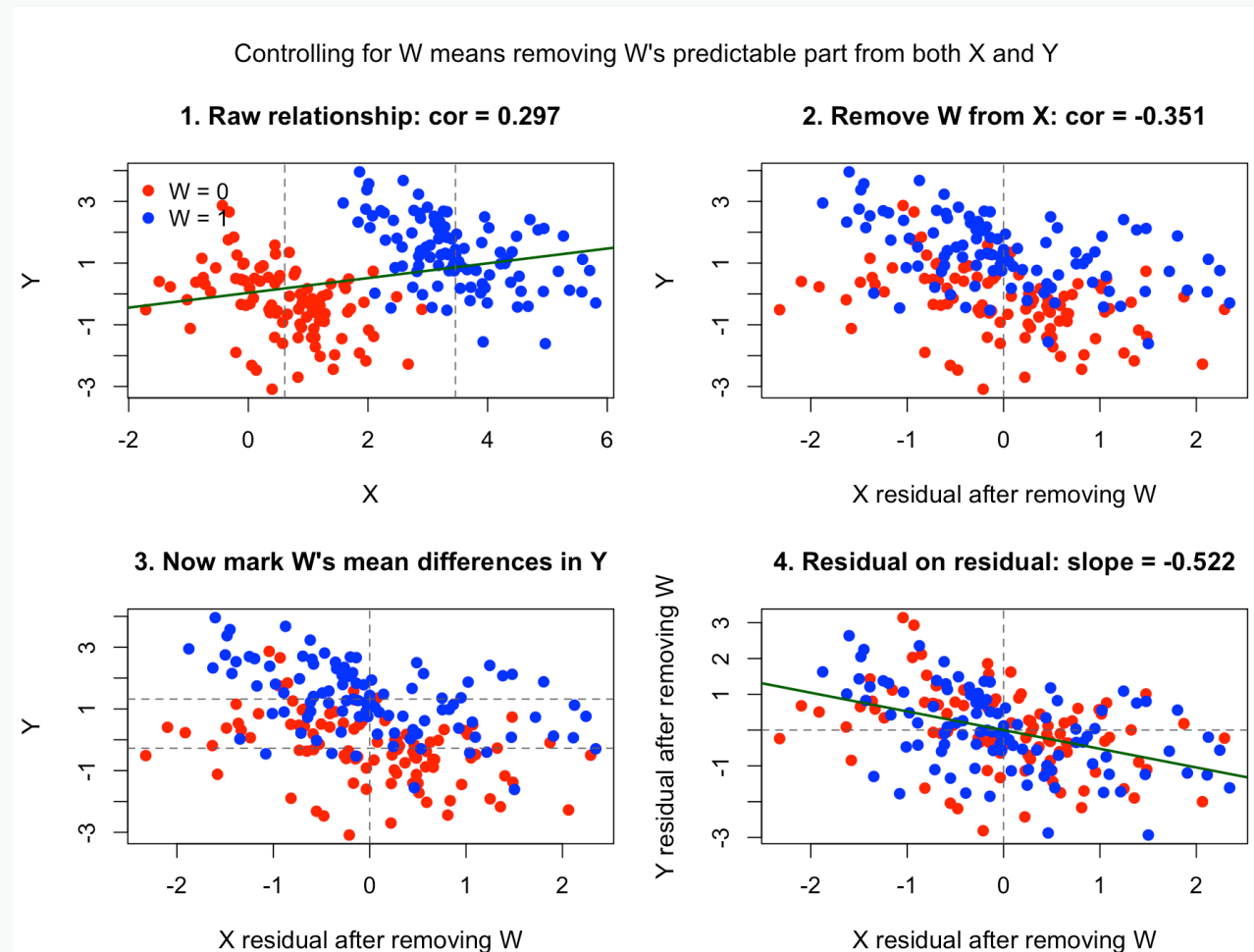
But the CAUSAL question is not:

| Do high-X observations also have high Y?

It could be:

| Within the same W context, does higher X predict lower Y?

Controlling means removing W from both sides



Regression is a conditional mean model.

FWL: Residuals recover the 'controlled slope'

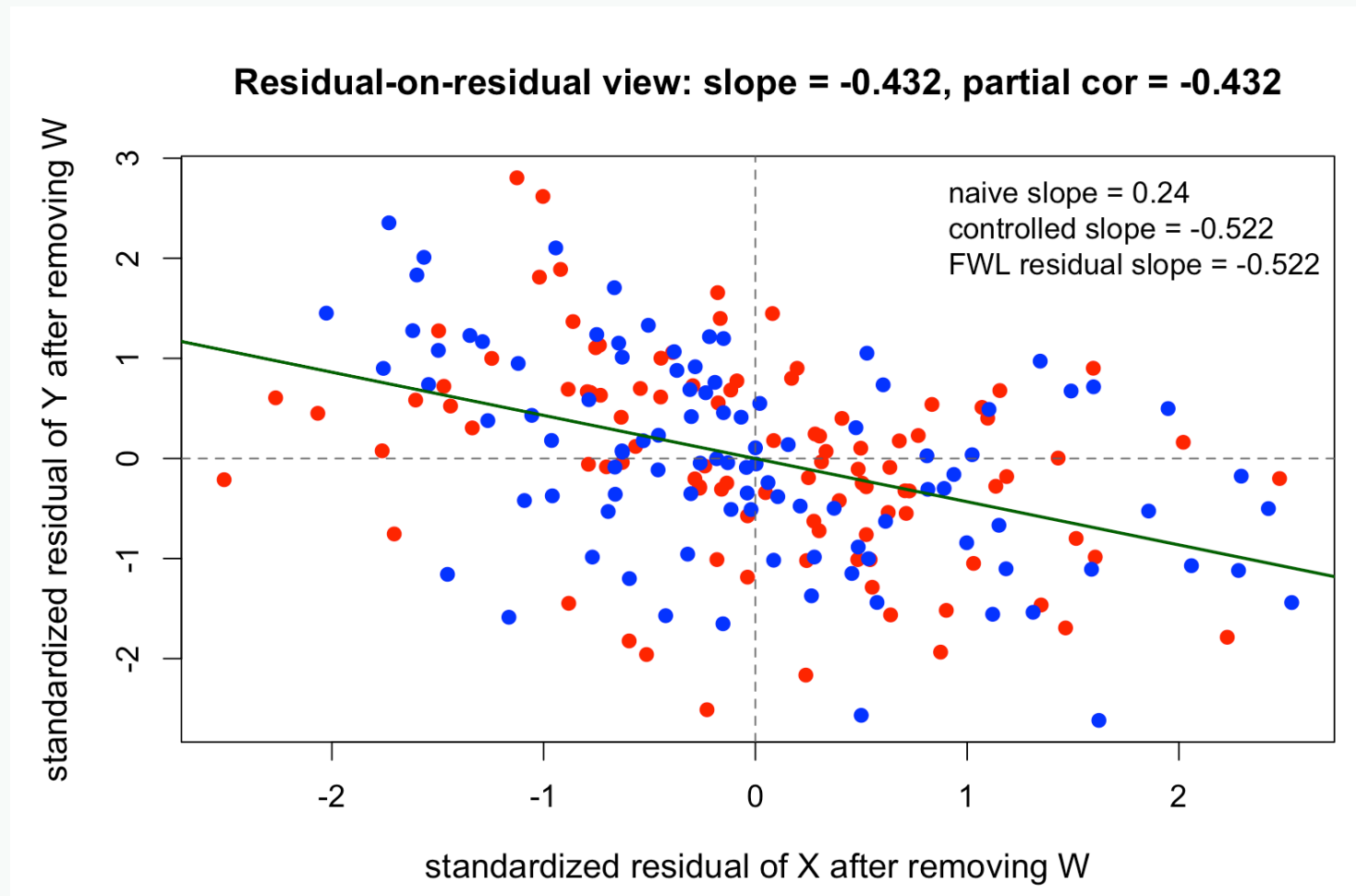
$$X_{\text{res}} = X - E[X \mid w]$$

$$Y_{\text{res}} = Y - E[Y \mid w]$$

controlled slope = slope from Y_{res} on X_{res}

This is the Frisch-Waugh-Lovell idea.

Residual plot reveals the DGP effect



Do you know the difference: partial correlation vs. regression coefficient?

DML 'scales' residualization

Ordinary control example

W is one simple control

learn $E[X_{\text{treat}} \text{ given } W]$ with OLS

learn $E[Y \text{ given } W]$ with OLS

regress residual Y on residual treatment

Causal ML version

X is a rich covariate set

learn $E[D \text{ given } X]$ with ML

learn $E[Y \text{ given } X]$ with ML

estimate an orthogonal residual comparison

DML = Double Machine Learning

Keep the notation straight

Teaching block	Focal variable	Context to remove	Residual comparison
Simple control	X	W	Y_res on X_res
Causal ML modules	D	rich X	Y_res on D_res

Same logic, different scale.

Module 1

Nonlinear **'observed'** selection

The confounders are observed, but the regression control list is too simple.

Start with the same residualization question

Suppose we want to estimate:

| Does treatment **D** change outcome **Y**?

The tempting regression is:

$$Y = \alpha + \beta D + \text{controls} + \text{error}$$

| But **D** was **NOT** randomly assigned.

DGP flow: observed context creates selection

Observed context X

$X \rightarrow \text{selection_score} = \text{nonlinear thresholds} + \text{interactions}$

$\text{selection_score} \rightarrow D = 1$ with probability $\text{logit}(\text{selection_score})$

$X \rightarrow \text{baseline_Y} = \text{nonlinear outcome risk}$

baseline_Y and $D \rightarrow Y = \text{baseline_Y} + 1.08D + \text{noise}$

The true treatment effect is constant.

Thinking step: what does "control for X" mean?

Regression-only thinking:

Add the observed covariates to the right-hand side.

Causal workflow thinking:

Ask how D was assigned as a function of X.
Ask how Y would vary with X even without D.

Old workflow confuses selection and effect

Estimate	Value
Simulated true effect	1.08
Naive regression	2.34
Linear controls	1.69
Hand-coded oracle controls	1.16

Linear controls move in the right direction, but do not solve the selection problem.

Why a CML-naive research keeps searching controls?

When the estimate changes after controls, the old workflow asks:

| Which control set gives the publishable coefficient?

The upgraded workflow asks:

| Which part of selection have we failed to model?

Note: CML = Causal Machine Learning

Causal ML move: learn the two parts to remove

$e(X) = P(D = 1 \mid X)$ removes X 's predictable part from D

$m(X) = E(Y \mid X)$ removes X 's predictable part from Y

These are not the causal effect.

They are the learned selection and baseline-risk maps.

We call them **nuisance**, b/c they are particularly taken care of by ML algorithms.

Neyman orthogonality protects the target

Small mistakes in nuisance models should not strongly distort the causal target.

This works only when the final score is built correctly.

Formal condition:

zero first-order Gateaux derivative

with respect to nuisance functions at their true values.

Estimate with the same residual logic

$$Y_{\text{res}} = Y - m_{\text{hat}}(X)$$

$$D_{\text{res}} = D - e_{\text{hat}}(X)$$

treatment effect = slope from Y_{res} on D_{res}

This is the intuition behind DML-style adjustment.

How the DML estimate is generated

Split the data into folds

Train $m_{\text{hat}}(X)$ and $e_{\text{hat}}(X)$ on the other folds

Predict Y and D for the held-out fold

Create Y_{res} and D_{res}

Regress Y_{res} on D_{res}

The last slope is the DML-style estimate.

How the doubly robust estimate is generated

Learn $e_hat(X)$ = treatment probability

Learn $\mu1_hat(X)$ = predicted Y if treated

Learn $\mu0_hat(X)$ = predicted Y if untreated

Build one corrected effect score per unit

Average the corrected scores

This produces the AIPW / DR row.

Doubly robust does not mean doubly magic

Outcome model	Propensity model	AIPW behavior
good	good	strongest case
good	bad	can still work
bad	good	can still work
bad	bad	fails

The protection is about observed covariates, not missing causes.

What are the two nuisance jobs?

Outcome nuisance:

$\mu_1(X)$, $\mu_0(X)$

Predict outcome risk under treatment and control.

Treatment nuisance:

$e(X)$

Predict who receives treatment.

'Neither' object is the causal effect.

Now the machine needs the brain's guidance.

Brain guidance	What the machine needs
Estimand	What effect are we estimating?
Learner choice	What estimated $e(X)$ and $\mu(X)$?
Cross-fitting	Were nuisance predictions out of fold?
Diagnostics	Did fit improve over linear benchmarks?
Design check	Are overlap and timing still credible?

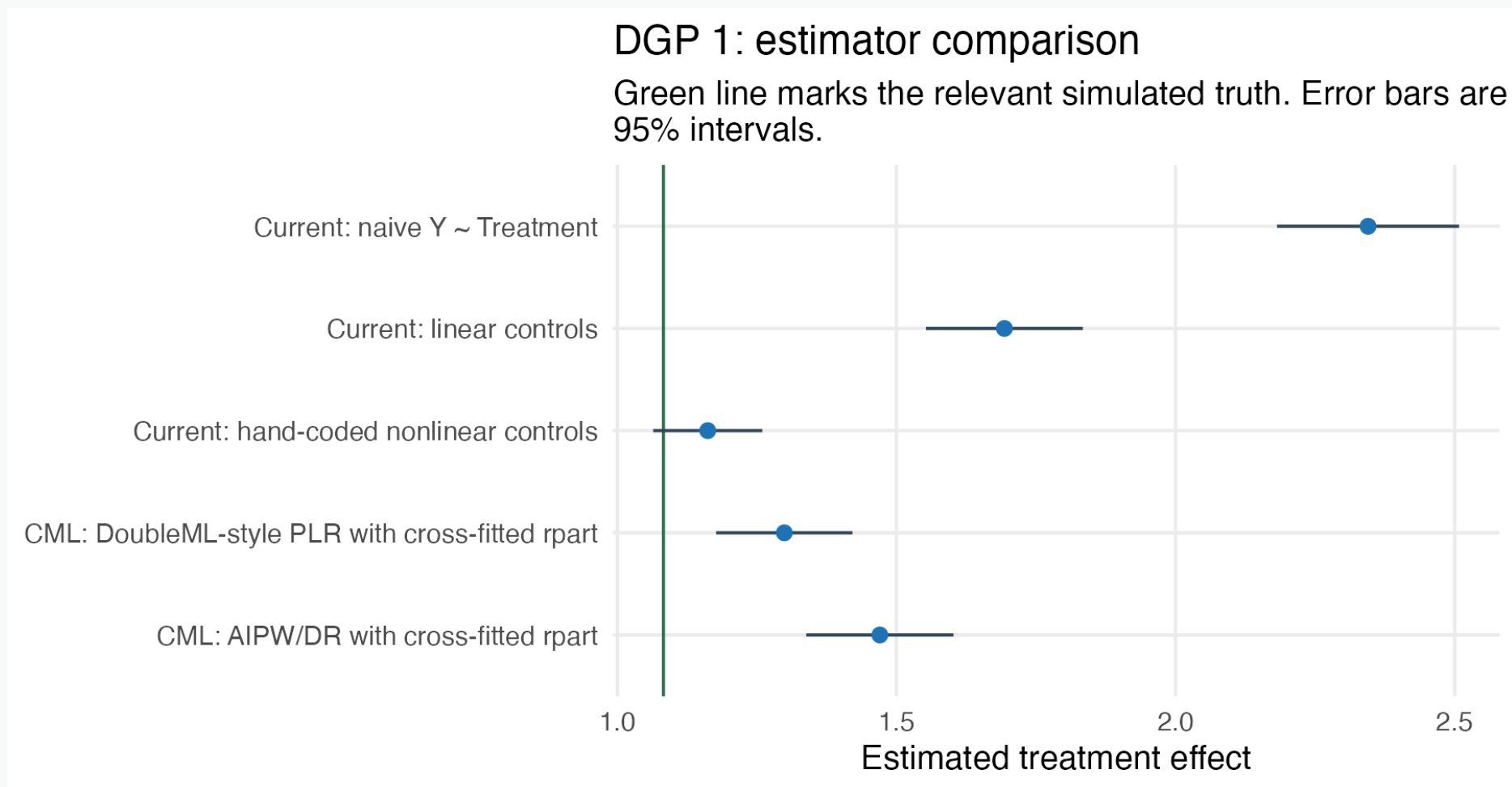
AIPW is a workflow step, not a final design by itself.

DML and AIPW solve different estimation jobs

Situation	Better starting point	Why
One focal coefficient	DML PLR	residualize Y and D , estimate one protected slope
Binary treatment ATE or ATT	AIPW / DR / IRM	combine outcome surfaces and propensity
One nuisance may be weak	DR score	can work if either outcome or propensity is right
Heterogeneity is central	DR/R/CATE learner	estimate effect variation after identification

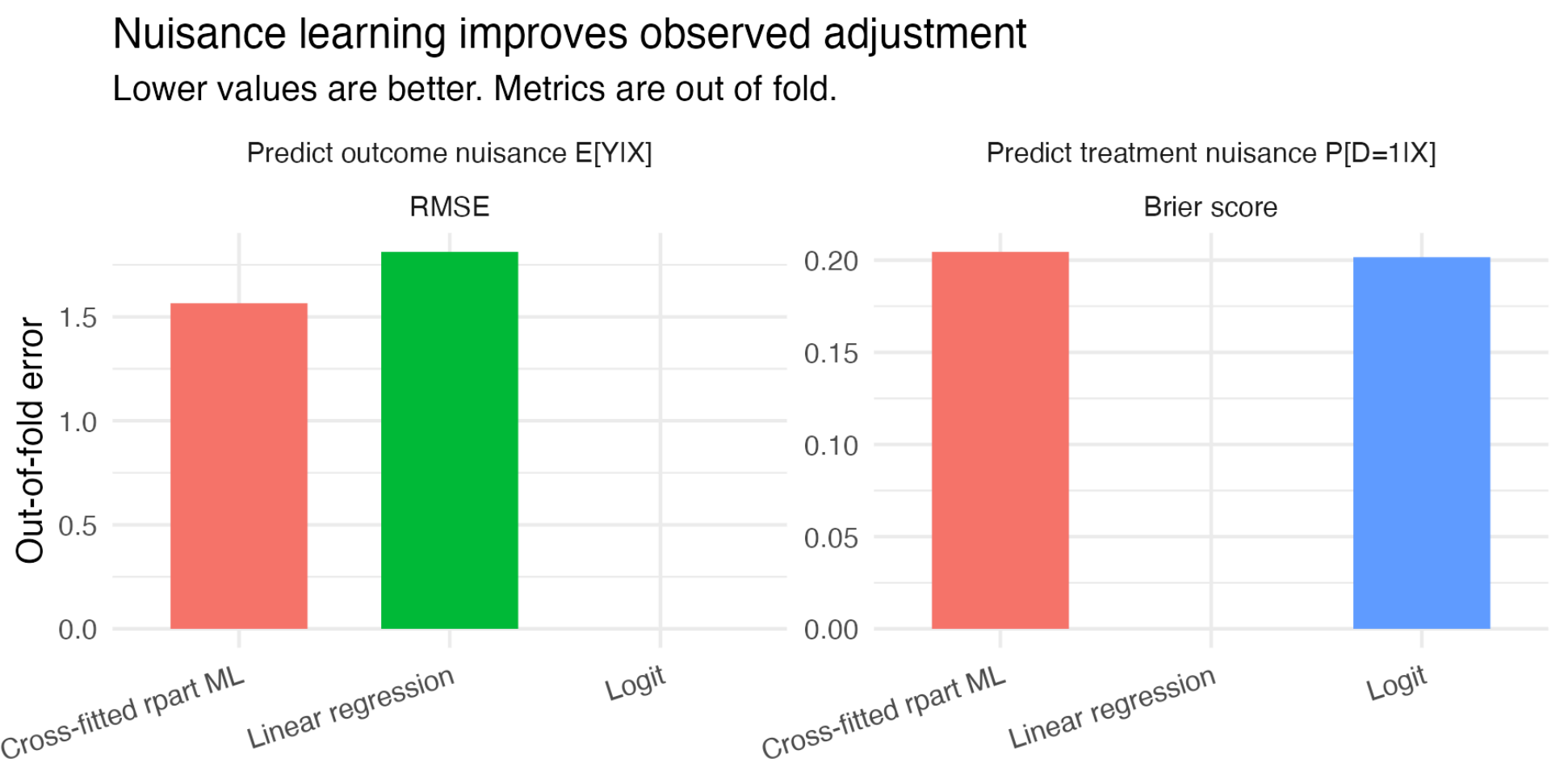
Note: ATE = average treatment effect: ATT = average treatment effect on the treated

What changed after the CML move?



Focus on the **bottom CML estimators**: DML and AIPW / DR.

Nuisance diagnostics show the learning



Module 1 upgrade

Replace:

| "Let me add more controls until beta behaves."

With:

| "Let me freeze OLS, learn selection and outcome risk out of fold, and **COMPARE** the estimators under the same estimand."

When estimates disagree, diagnose the pattern

Pattern	What it suggests
OLS weak, DML clearer	Linear controls missed adjustment structure
OLS clear, DML weak	OLS result may reflect selection
OLS and DML both weak	Average effect is not supported yet
ATE weak, CATE clear	Effect may exist only in some contexts

The pattern tells you the next theory move

Pattern	Diagnosis	Theory move
OLS weak, DML clearer	Adjustment	Theory may survive stronger adjustment
OLS clear, DML weak	Selection	Reframe as selection or description
OLS and DML both weak	Recheck	Check timing, measures, overlap, or HTE
ATE weak, CATE clear	Boundary	Build boundary-condition theory

What Module 1 should teach us

Hidden confounding: **observed** selection can still be **missing** a cause

Causal ML can learn complex observed selection.

It cannot learn a cause that is missing from the data.

DGP flow: a missing common cause

Observed context $X \rightarrow D$

Observed context $X \rightarrow Y$

Unobserved $U \rightarrow D$

Unobserved $U \rightarrow Y$

$D \rightarrow Y$

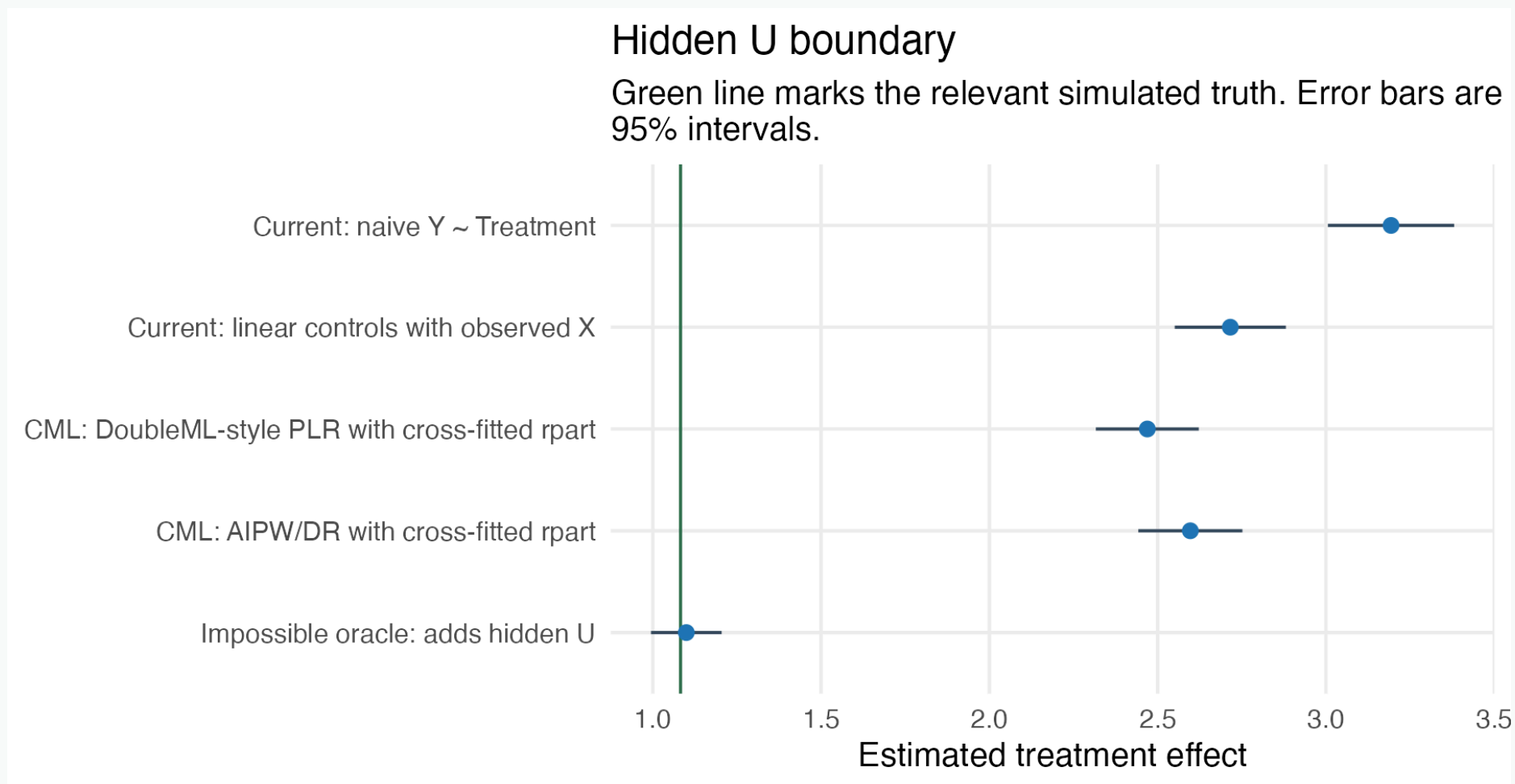
The researcher sends X to the learner, but U is not in the feature library.

Bias can still look disciplined

Model	What the analyst may think	What is still wrong
$Y \sim D$	raw effect	confounded by U
$Y \sim D + X$	controlled effect	U still omitted
$Y \sim D + \text{many } X$	robust effect	U still omitted
DML/AIPW with X	flexible adjustment	U still omitted

The problem is not flexibility. It is **missing identification**.

Scoreboard: ML still misses the truth



Focus on the **third and fourth estimators**: both CML rows still miss U.

Module 1 upgrade: hidden confounding

Replace:

| "Maybe a better algorithm will fix this."

With:

| "If the common cause is unobserved, I need design, new data, sensitivity analysis, or a weaker claim."

Boundary Module

Sensitivity and refutation (and some epistemology, perhaps)

When hidden confounding is plausible, the next move is **not** another model.

It is a claim-strength audit.

After estimation, ask how fragile the claim is

Causal ML estimate

|

v

Refutation and sensitivity

|

v

Claim-strength decision

The estimate is one input to the claim, not the claim itself.

Four refutation checks

Check	Question
Placebo outcome	Does D predict an outcome it should not affect?
Placebo treatment	Does fake D predict Y?
Negative control	Is there evidence of residual selection?
Omitted-variable sensitivity	How strong must hidden U be to explain the result?



Emily Oster

Match package to job

Job	Tool family
Graph assumptions and run refuters	DoWhy
Omitted-variable sensitivity	sensemakr / Oster-style logic
Rich observed-selection benchmark	ML nuisance models
Final adjusted estimate	DoubleML / AIPW

Sensitivity is not a backup estimator.

Claim decision after sensitivity

Pattern	Claim to write
Estimate stable, placebo clean, overlap good	stronger causal claim
Estimate stable but sensitivity fragile	cautious causal claim
Placebo or negative control fails	likely residual confounding
Hidden U could easily explain effect	descriptive claim or redesign

The workflow upgrades the claim, not only the coefficient.

Module 2

Overlap and Support: Welcome to counterfactuals!

A coefficient can exist even when the comparison is weak.

DGP flow: selection becomes **too strong**

Observed context X

$X \rightarrow \text{selection_score} = \text{stronger nonlinear selection rule}$

$\text{selection_score} \rightarrow e(X)$ close to 0 or 1 for some units

$e(X) \rightarrow D$

$X \text{ and } D \rightarrow Y$

The regression still returns a coefficient, but some counterfactual comparisons are **thin**.

Heraclitus: "We both step and do not step in the same rivers. We are and are not."

Adjustment needs comparison units

Before asking whether the effect is significant, ask:

Region of X	What do we need?
Treated units appear	Comparable untreated units
Untreated units appear	Comparable treated units
One group is missing	Redefine or restrict the estimand

without reliable counterfactuals: Tables hide support

Estimate	Value
Simulated true effect	1.08
Naive regression	2.79
Linear controls	1.96
AIPW full sample	1.71

The problem is not only model flexibility. It is the comparison set.

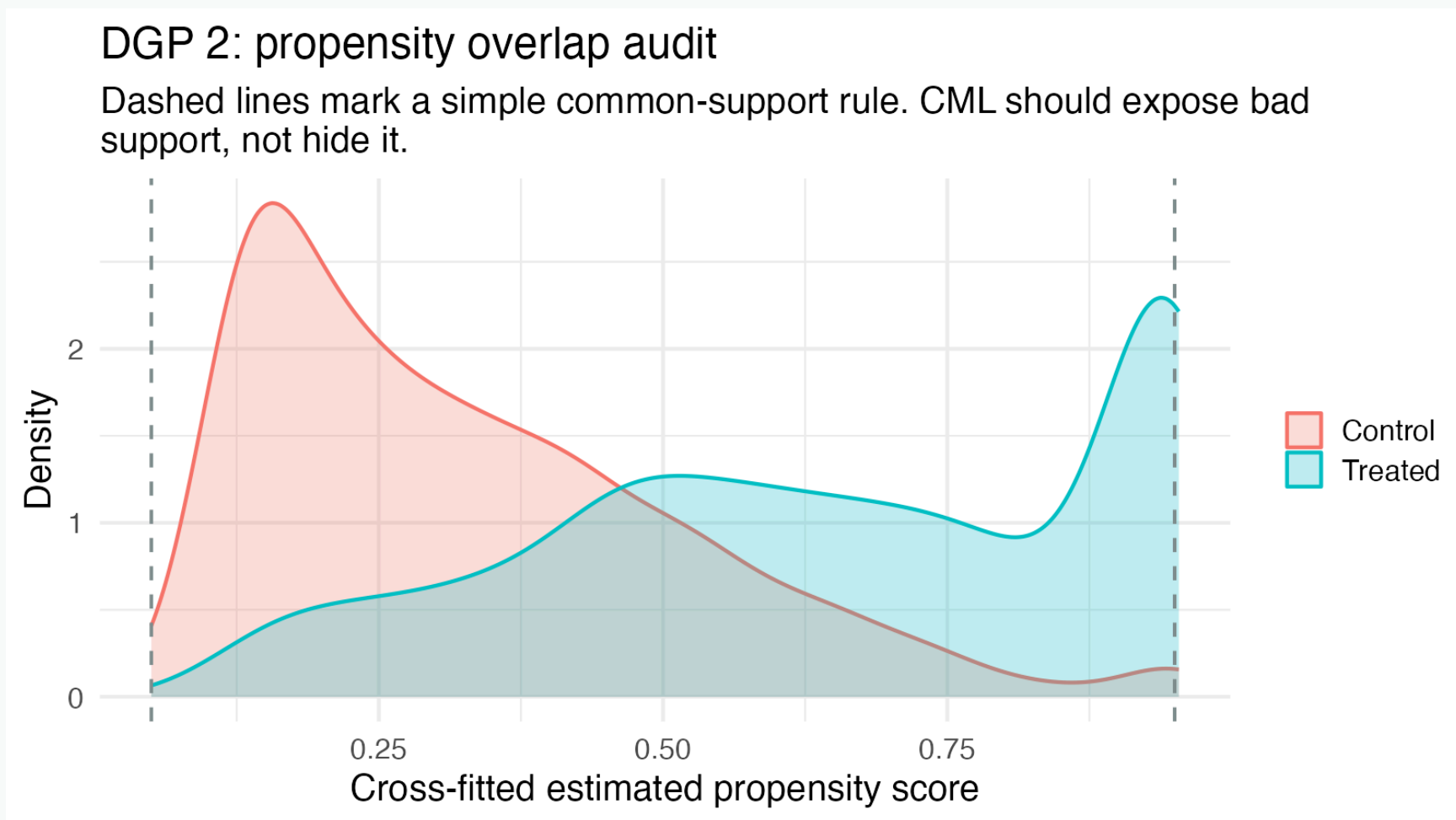
Causal ML move: use propensity as a diagnostic

Learn $e(X) = P(D = 1 \mid X)$

Then inspect:

- Do treated and untreated units overlap?
- Are weights concentrated?
- Which population is actually comparable?

The overlap plot changes the next question



How the overlap numbers are generated

Fit cross-fitted $e_{\text{hat}}(X)$

Compute IPW weight for each unit

Summarize extreme propensities

Compute effective sample size

Choose the estimand the data can support

Support diagnostics force a design decision

Diagnostic	Value
Nominal sample size	2600
Effective sample size after weighting	1329
Share with <code>e_hat > .95</code>	3.3%
Maximum weight	21.56

Restrict the estimand (if needed?)

Full-sample ATE:

What is the effect for everyone?

Weak overlap:

Some units lack credible counterfactuals.

Restriction:

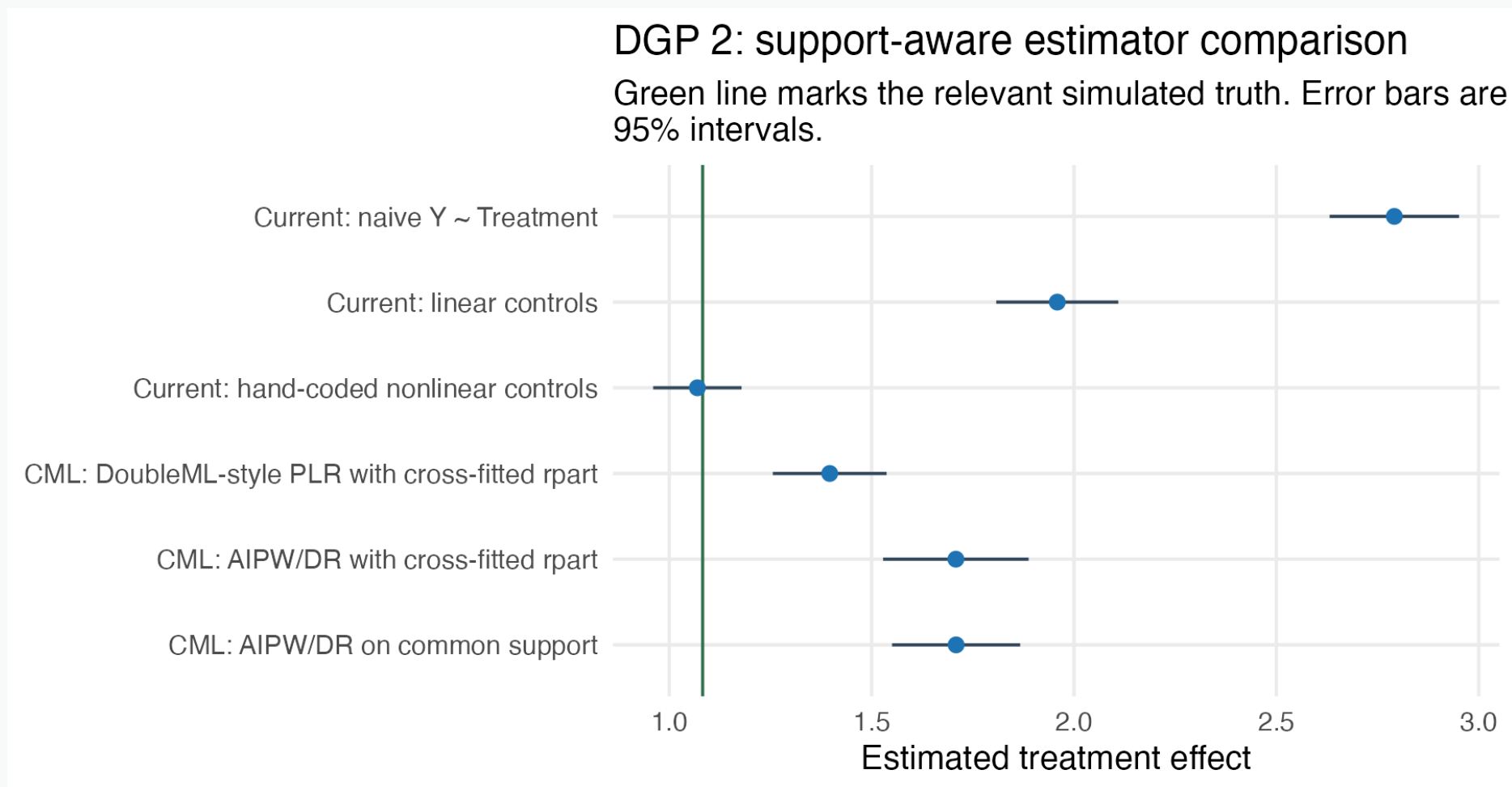
$\text{keep } .05 \leq e_{\text{hat}} \leq .95$

New target:

common-support ATE

New Target: common-support ATE

The scoreboard should not hide bad support



Focus on the **bottom three CML estimators**.

Module 2 upgrade

Replace:

| "I controlled for X , so the estimate is adjusted."

With:

| "I checked whether treated and untreated units **overlap**
in X , then chose the estimand that the data can **support**."

Module 3

Bad controls and estimand drift

A **better-looking** coefficient can answer the **wrong** question.

DGP flow: treatment creates a mediator

Observed context $X \rightarrow \text{baseline}_Y$

$D \rightarrow \text{post_attention} = 0.95D + \text{context} + \text{noise}$

$D \rightarrow \text{direct } Y \text{ gain} = 0.75D$

$\text{post_attention} \rightarrow \text{mediated } Y \text{ gain} = 0.35 * \text{post_attention}$

$Y = \text{baseline}_Y + \text{direct gain} + \text{mediated gain} + \text{noise}$

Total effect is about $0.75 + 0.95 * 0.35 = 1.08$.

Why the mediator changes the question

Without post_attention:

D carries direct path + mediated path

With post_attention:

D mostly carries the direct path only

The coefficient can look cleaner while the estimand changes.

The old workflow sees a **'useful'** control

Why add `post_attention`?

- It predicts the outcome.
- It improves model fit.
- It changes the treatment coefficient.
- It seems substantively meaningful.

All four can be true, and the control can still be **wrong** for a total-effect estimand.

Thinking step: controls must have roles

Variable role	Use for total effect?
Pre-treatment confounder	Yes
Baseline outcome or pretrend	Yes
Precision covariate	Yes, if pre-treatment
Mediator	No
Collider or selection variable	No
Post-treatment variable	No for total effect

Filter features before learning

The workflow is:

1. Write the estimand.
2. Assign each variable a timing.
3. Assign each variable a causal role.
4. Send only allowed variables to DML, AIPW, or CATE tools.

How the bad-control plot is generated

Generate data with a mediator

Estimate total-effect models without the mediator

Estimate one model that adds `post_attention`

Compare estimates to the relevant truth

The mediator model asks a different question

Models without post_attention:

$D \rightarrow Y$

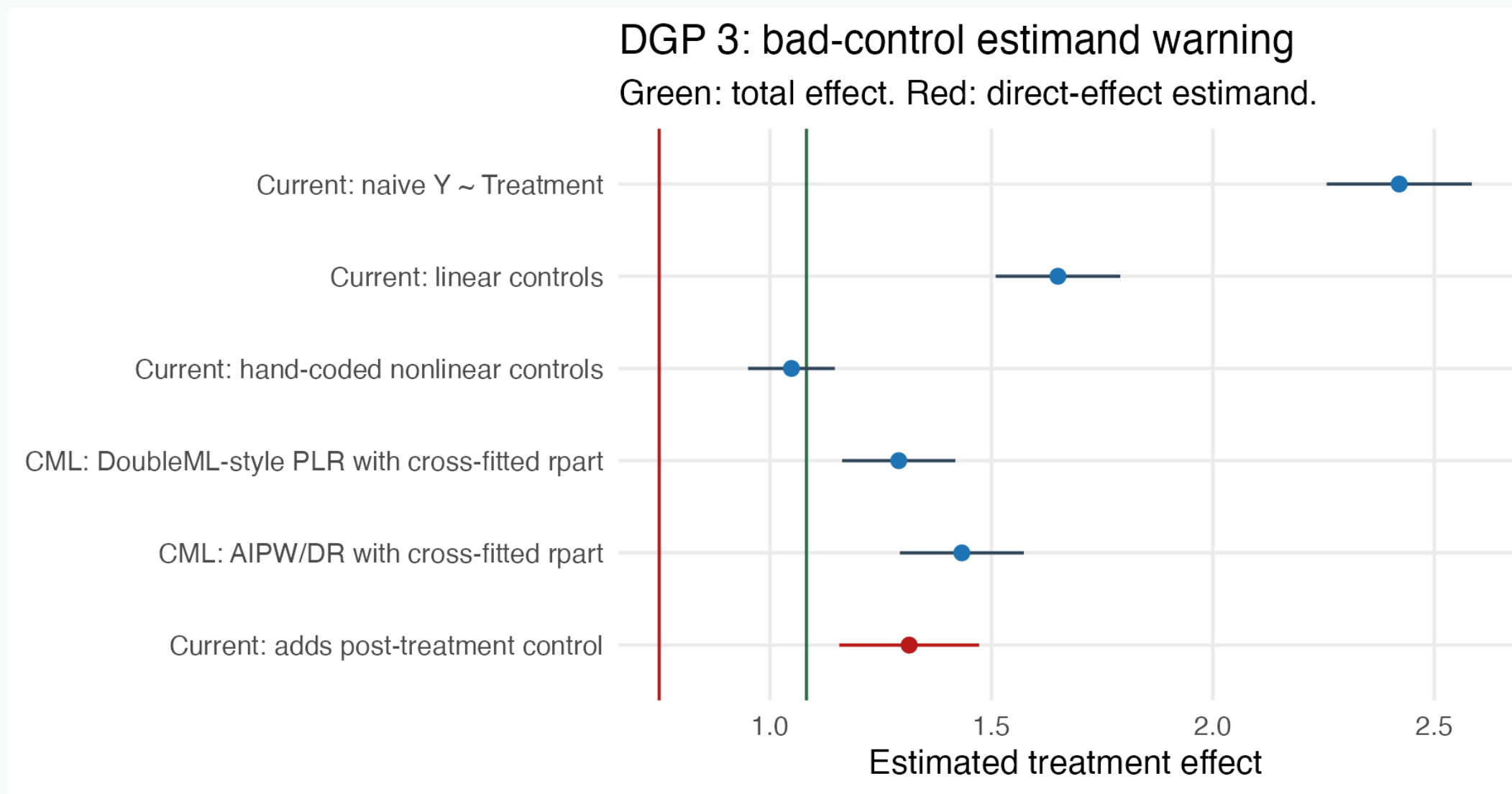
$D \rightarrow \text{post_attention} \rightarrow Y$

Model with post_attention:

$D \rightarrow Y$, holding post_attention fixed

Same data, different estimand.

The bad-control plot is an estimand warning



Module 3 upgrade

Replace:

| "This variable predicts Y, so I should control for it."

With:

| "This variable must be pre-treatment and **causally allowed** for the estimand I am estimating."

Charlie Munger: "I am personally skeptical of some of the hype that has gone into artificial intelligence. I think old-fashioned intelligence works pretty well."

Module 4

Heterogeneous treatment effects

Moderators should be **validated** boundary conditions, not discoveries from one table.

DGP flow: treatment effects vary by context

Observed context X

$X \rightarrow$ subgroup features:

high digital growth, turnaround, high debt, visible R&D

subgroup features $\rightarrow \tau_i$

$Y = \text{baseline}_Y + \tau_i * D + \text{noise}$

The moderator may be a region of X , not one variable.

Key object: **τ_i**

The old workflow hunts interactions

```
Y ~ D * moderator_1 + controls  
Y ~ D * moderator_2 + controls  
Y ~ D * moderator_3 + controls  
...
```

The first significant and plausible term **luckily and sadly** becomes theory.

Separate discovery from validation

Job	Regression-only habit	Upgraded habit
Discover	screen many interactions	learn CATE pattern
Validate	same coefficient table	holdout or cross-fit
Theorize	explain the winner	explain stable effect groups

A brainless machine (ML or regression) can mass-produces 'delusion'.

CATE first predicts unit-level effects

```
Firm A -> predicted effect = 0.3  
Firm B -> predicted effect = 1.1  
Firm C -> predicted effect = 2.4  
...
```

Then we group units to make the pattern meaningful (meta-learners).

A T-learner uses two outcome models

Model among treated units:

$\mu_1\text{-hat}(X)$ = predicted Y if treated

Model among control units:

$\mu_0\text{-hat}(X)$ = predicted Y if untreated

$\text{CATE}_i = \mu_1\text{-hat}(X_i) - \mu_0\text{-hat}(X_i)$

Remember: They differ in the "architecture"--they structure and "sequence the steps" to predict counterfactual outcomes.

An R-learner extends the DML logic

First clean Y of X .

First clean D of X .

Then learn whether the cleaned slope varies by X .

DML estimates one average cleaned slope.

R-learner asks whether that slope differs by unit type.

Meta-learners give different moderator clues

Learner	Core idea	Moderator use
S-learner	one outcome model	quick screen
T-learner	treated vs control models	intuitive CATE contrast
X-learner	impute effects, then blend	unbalanced groups
R/DR-learner	orthogonal CATE scores	selection-adjusted heterogeneity

Use meta-learners to form moderator hypotheses

Predict $CATE_i$ for each unit

Rank units by predicted effect

Profile high-effect versus low-effect units

Name the pre-treatment pattern

Validate the pattern in holdout data

Groups turn CATE scores into moderator clues

Sort units by predicted effect

Bottom third -> low-effect group

Middle third -> middle-effect group

Top third -> high-effect group

Then inspect **what distinguishes** the groups.

In the code: Locate features where tau_hat varies

Compare profiles	Low tau_hat units	High tau_hat units
Digital growth	lower	higher
Turnaround context	rare	common
Debt pressure	higher	lower
R&D visibility	weaker	stronger

Candidate moderator: growth and turnaround capacity.

Validate before claiming moderation

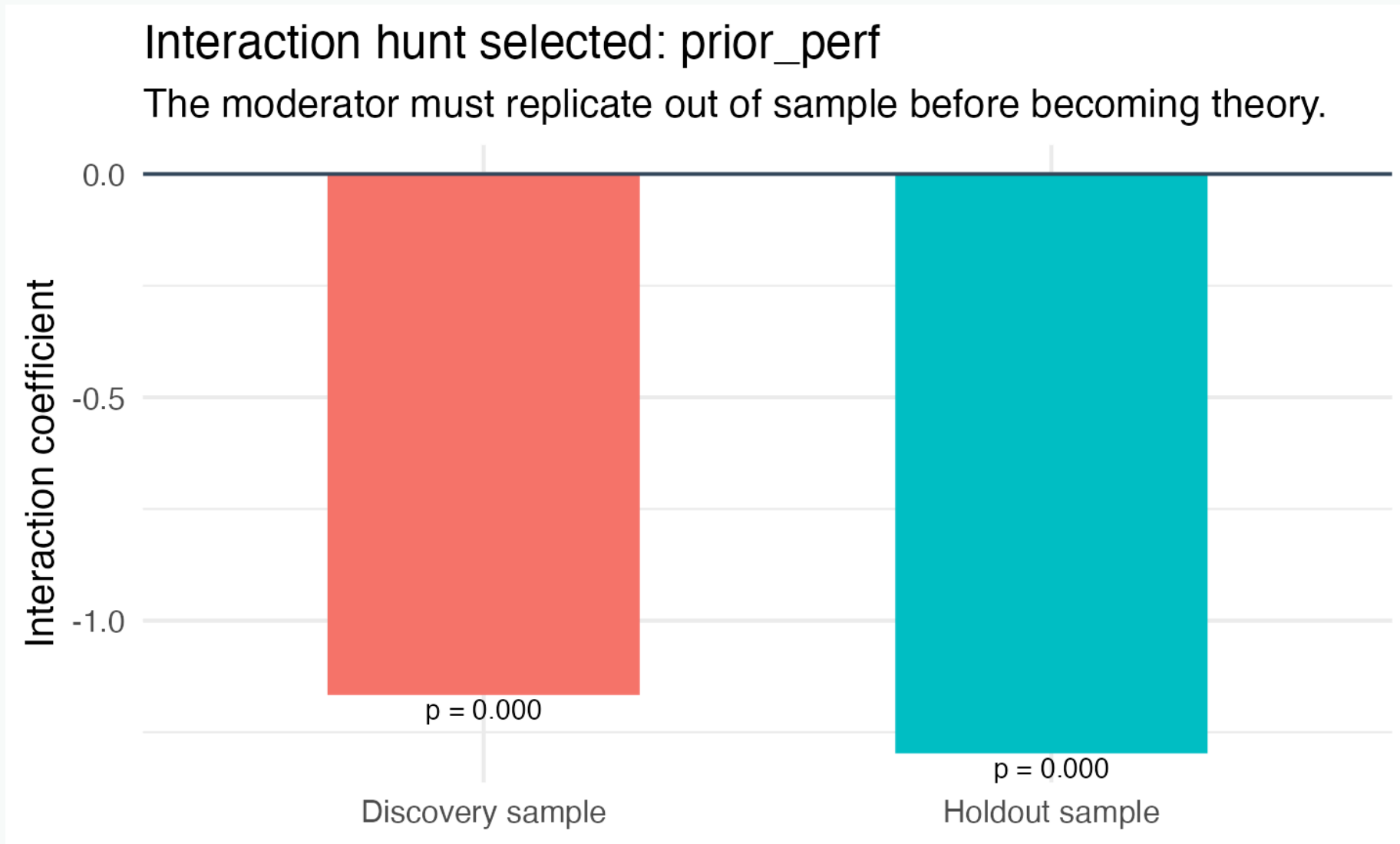
Discovery sample:
learn the CATE pattern

Holdout sample:
test whether the same pattern predicts stronger effects

Only stable patterns become theory

Only **stable** patterns become theory.

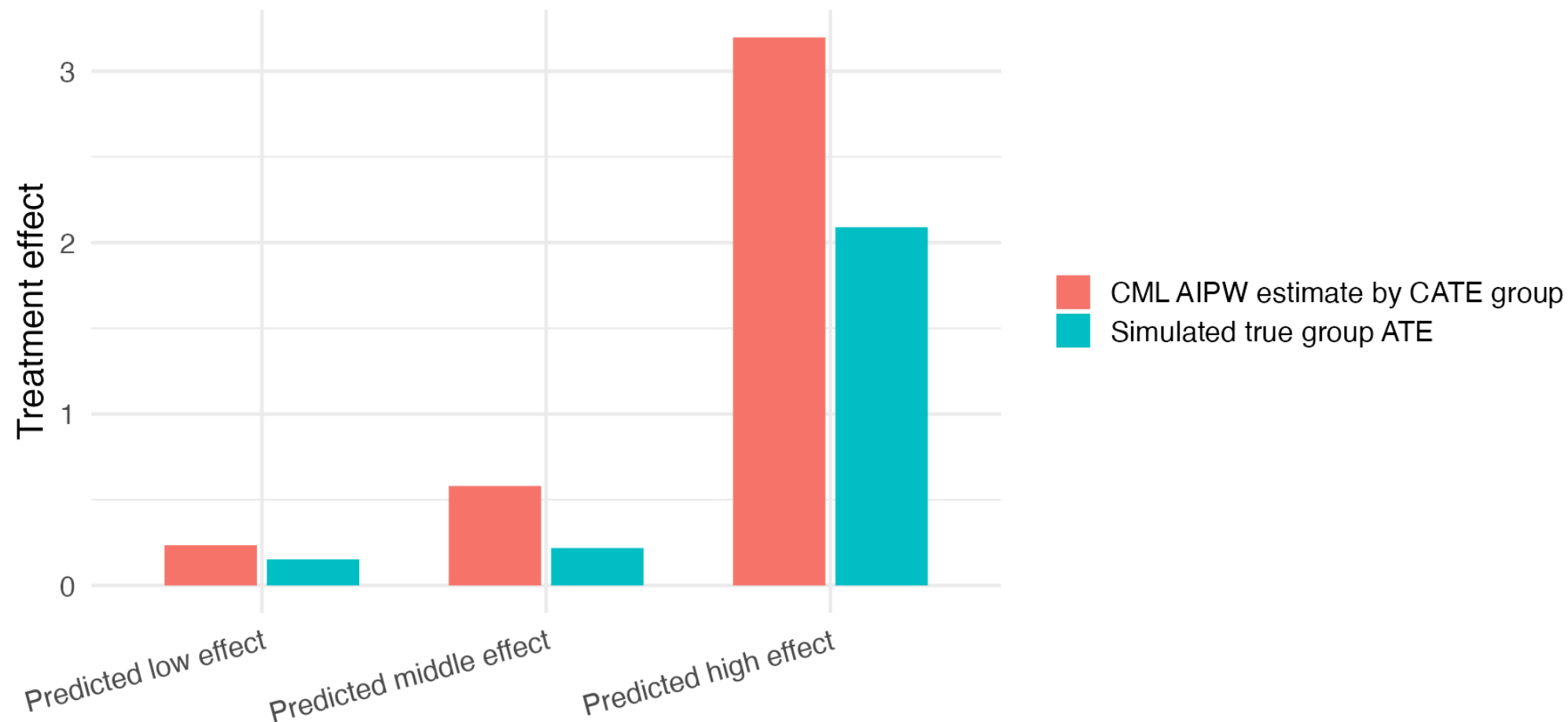
Holdout checks separate discovery from theory



Validate that CATE groups really differ

Validate heterogeneity before theory

Estimated CATE groups are compared with simulated ground truth.



Module 4 upgrade

Replace:

| "The significant interaction is my moderator."

With:

| "The **validated** CATE pattern tells me **where** the treatment works better or worse, and only then do I build a moderator theory."

The package map should follow the **workflow** **gate**

Job	Package family
Graph, identify, refute	DoWhy
Sensitivity and placebo checks	DoWhy refuters, sensemakr, Oster-style logic
Orthogonal ATE or ATT	DoubleML
CATE and policy learning	EconML, CausalML, grf
Staggered adoption	did
Regression benchmark	fixest, PyFixest, statsmodels

Have you tried 'AI-agents' (); create your own workflows.

The paper workflow becomes a sequence of gates

Before this claim	Show this diagnostic
Main effect	estimand, variable roles, frozen OLS, DML or DR comparison
Adjusted effect	nuisance diagnostics, hidden-confounding boundary, and overlap
Causal interpretation	refutation and omitted-variable sensitivity
Moderator	held-out CATE or interaction validation
Conclusion	remaining threats: hidden confounding, weak support, timing

When regression stalls

Do not add controls first. Choose the **diagnostic** branch.

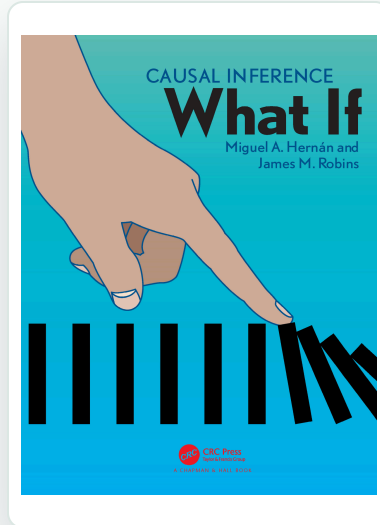
Symptom	First diagnostic move
Estimand unclear	rewrite the intervention question
Controls move the coefficient	check observed selection and nuisance fit
OLS clear, DML weak	treat OLS as possible selection or misspecification
Overlap weak	restrict estimand, redesign sample, or stop

If diagnostics stay weak

Move the claim only after naming the failure mode.

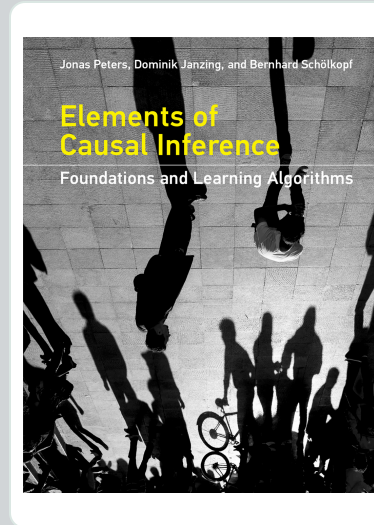
Symptom	First diagnostic move
Predictive control is post-treatment	rebuild the feature library
Hidden U plausible	sensitivity, negative controls, or design repair
ATE weak, CATE stable	build boundary-condition theory
All paths weak	repair measurement or report descriptive evidence

Reference Books



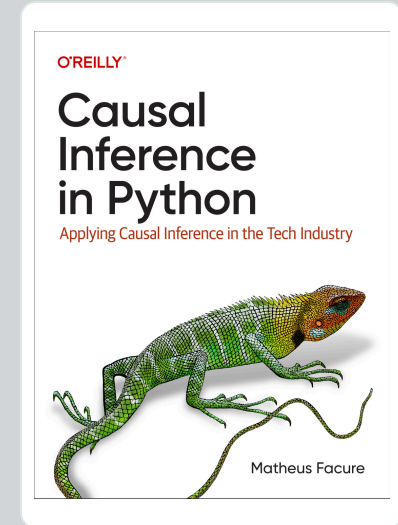
What If

Miguel A. Hernan and
James M. Robins



Elements of Causal Inference

Jonas Peters, Dominik
Janzing, and Bernhard
Scholkopf



Causal Inference in Python

Matheus Facure